

DSC8005: OPTIMIZATION

Effective Term

Semester A 2026/27

Part I Course Overview

Course Title

Optimization

Subject Code

DSC - Data Science

Course Number

8005

Academic Unit

Data Science (DS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

R8 - Research Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

SDSC8005 Optimization

Exclusive Courses

Nil

Additional Information

Nil

Part II Course Details

Abstract

In this course we will learn how to formulate, analyze and solve various optimization problems. Topics include convex analysis, classifying different types of optimization problems optimality conditions duality, unconstrained optimization, and optimization under uncertainty. No prior optimization background is required for this class. However, students should have workable knowledge in multivariable calculus, linear algebra and matrix theory.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if app.)	DEC-A1	DEC-A2	DEC-A3
1	Explain the fundamentals and principles of optimization	20	x	x	
2	Explain the theory for different types of optimization problems	50		x	x
3	Use/apply software packages or existing algorithms to formulate and solve various optimization problems	15		x	
4	Model different applications of optimization	15	x	x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Class Activities	Students will engage in lectures which introduce the understanding and applications of various concepts and methods.	1, 2, 3, 4	32 hours/sem
2	Tutorial Sessions	Students will participate tutorial sessions that have explanation and exercises to familiarize themselves with the methods learnt during the lectures.	1, 2, 3, 4	7 hours/sem

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- for nil entry)	Allow Use of GenAI?
1	Assignments Students are required to explain knowledge and apply methodologies learned from the course in solving some problems.	1, 2, 3, 4	20	-	Yes
2	Midterm/in-lab/take-home exam Students will be assessed via the examination in their ability to explain concepts and apply methodologies in solving optimization problems	1, 2, 3, 4	50	-	No
3	Course project Students will be assessed via course projects in demonstrating their ability in solving real optimization problems	2, 3, 4	30	-	Yes

Continuous Assessment (%)

100

Examination (%)

0

Assessment Rubrics (AR)**Assessment Task**

Assignments (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Train students' ability to theorize and solve various types of optimization problems.

Excellent

A+, A, A-

Good

B+, B, B-

Fair

C+, C, C-

Marginal

D

Failure

F

Assessment Task

Midterm/in-lab/take-home exam (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

It assesses students' ability to theorize and solve various types of optimization problems.

Excellent

A+, A, A-

Good

B+, B, B-

Fair

C+, C, C-

Marginal

D

Failure

F

Assessment Task

Course project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Nurture students' ability in solving real optimization problems.

Excellent

A+, A, A-

Good

B+, B, B-

Fair

C+, C, C-

Marginal

D

Failure

F

Assessment Task

Assignments (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Train students' ability to theorize and solve various types of optimization problems.

Excellent

A+, A, A-

Good

B+, B

Marginal

B-, C+, C

Failure

F

Assessment Task

Midterm/in-lab/take-home exam (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

It assesses students' ability to theorize and solve various types of optimization problems.

Excellent

A+, A, A-

Good

B+, B

Marginal

B-, C+, C

Failure

F

Assessment Task

Course project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

Nurture students' ability in solving real optimization problems.

Excellent

A+, A, A-

Good

B+, B

Marginal

B-, C+, C

Failure

F

Part III Other Information

Keyword Syllabus

Convex analysis, optimality conditions duality, unconstrained optimization, optimization under uncertainty

Reading List

Compulsory Readings

Title	
1	S. Boyd and L. Vandenberghe. Convex Optimization. Cambridge University Press, Cambridge, 2004. Available online at http://www.stanford.edu/~boyd/cvxbook/ .
2	Lecture Notes and Slides

Additional Readings

Title	
1	A. Ben-Tal and A. Nemirovski. Lectures on Modern Convex Optimization: Analysis, Algorithms, and Engineering Applications, volume 2 of MPS{SIAM Series on Optimization. Society for Industrial and Applied Mathematics, Philadelphia, Pennsylvania, 2001.
2	M. S. Bazaraa, H. D. Sherali, and C. M. Shetty. Nonlinear Programming: Theory and Algorithms. Wiley{Interscience Series in Discrete Mathematics and Optimization. John Wiley & Sons, Inc., New York, second edition, 1993.
3	D. P. Bertsekas. Nonlinear Programming. Athena Scientific, Belmont, Massachusetts, second edition, 1999.
4	D. G. Luenberger and Y. Ye. Linear and Nonlinear Programming, volume 116 of International Series in Operations Research and Management Science. Springer Science+Business Media, LLC, New York, third edition, 2008.